Naman Chetwani

+1 925 895 9674 | naman.chetwani@gmail.com | github.com/coder-2011 | naman.world

PROJECTS

wrdn

- Building **wrdn**, an accountability tool that uses audio and visual context to track how time is spent and compare it against a user's stated goals.
- Designed the memory layer, agent command execution, and hardware plan across a clip-on device, Meta Ray-Bans, and Brilliant Labs Halo.
- Building the stack **from kernels to hardware**, with inspectable code for users who want to verify how the system works.

Gemma 4 Inference Megakernel

- Building **gemma.c**, an experimental Gemma 4 dense inference runtime: standalone CUDA/PTX kernels first, then a correct unfused baseline and measured fusion into larger kernels.

STRATUS UAV: Optimizing Prescribed Burning using UAVs and Predictive Modeling

- Led development of a **patent-pending autonomous UAV** to mitigate wildfire spread; 3D-printed a VTOL aircraft for autonomous backburning with **15 km range** and **1.2 hour endurance**.
- Targeted core field constraints: helicopter operations around $16k/day, custom aircraft and pilots, and pilot liability; designed for **5x lower price** and **3x higher range** than the closest drone competitor.

SBON (Strategic-Blackline-Optimization-Engine)

- Built a wildfire prediction model that **set state-of-the-art results** and created the first ML approach to optimizing prescribed burns.
- Built a tri-branch multimodal architecture across **13 input modalities** spanning fire imagery, weather, and geography; ConvNeXt and SSM encoders feed a larger SSM for 4-hour fire-mask prediction.
- Treated the model as an environment, rolled out predictions to **48 hours**, and used PPO to optimize burn placement in a niche field with fewer than 20 papers in the last 30 years.

The no-circles project — no-circles.com

- Built a personalized daily newsletter that learns from each reader's replies and behavior; scaled to **triple-digit users within 3 weeks**.
- Built the user memory, daily brief, and reply feedback loop to explain unfamiliar fields through topics each reader already cares about.

PHOENIX UAV / Science Fair Results

- Built PHOENIX UAV for early wildfire detection; placed **2nd in the California State Science Fair**, **1st at regionals**, and won 5 awards including the Henkel aerospace award and Junior Lemelson-MIT Inventor Award.

WORK

Portfolio Manager

- Manage a **seven-figure household equity portfolio**, up **47% YTD** as of May 31, 2026; trades are partially handled through software agents that monitor markets, flag changes, and execute position updates.

Fire Technology Contract Work

Hardware / Systems Engineer

- Under contract building machines for fire tech companies, combining autonomous hardware, manufacturing, and systems built for field use.

EDUCATION

Independent Study / Competitions

- Selected for **SPARC '26** (3% acceptance rate); achieved **USACO Gold** and F=MA qualifying scores in the top 5% nationally.
- Competed in CALICO, Harker Physics Invitational, and Harker Programming Invitational; held a top 30 national placement in MSPF Debate with 2 tournament wins and elimination rounds at Stanford and TOC.